

Robustness of Treatment Effects between Common and Heterogeneous Treatment Effects

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Abstract

Empirical evaluations of policy interventions routinely interpret causal effects as homogeneous across individuals. When outcomes are discrete, however, this assumption can both mislead prediction and misrepresent the policy-relevant treatment effect. This paper (i) separates the common and differential components of a treatment effect, (ii) characterizes the identified set for the differential component using Fréchet–Hoeffding bounds and partial-identification arguments, and (iii) assesses the robustness of estimated treatment effects to departures from the common-effects assumption.

1 Introduction

Empirical work on policy interventions often interprets the causal effect of a treatment as homogeneous across individuals i and j . In settings where the outcome is not continuous, this practice carries two costs: it generates systematic prediction error, and it can fail to represent the treatment effect that is actually relevant for policy. The object that an analyst reports under a common-effects assumption need not coincide with the distribution of effects in the population it is meant to describe. This paper makes three contributions. First, it separates the common and differential components of a treatment effect, making explicit the part of the effect that is shared across units and the part that varies. Second, it characterizes the differential component through Fréchet–Hoeffding bounds and partial identification, recovering the identified set rather than relying on an assumption that point-identifies a single number. Third, it documents the robustness of estimated treatment effects as the analysis moves from the common-effects benchmark to a model that admits heterogeneity.

The remainder of the paper is organized as follows. Section 2 sets out the identification framework — randomization identifies the two marginal distributions of the potential outcomes but not their joint law — and derives from a single proposition three canonical cases, in which heterogeneity is undetermined, forced by the data, or absent. Section 3 then sets up the potential-outcomes framework, defines the common and differential effects, and positions four parameters for impact heterogeneity. Section 4 develops the partial-identification results using Fréchet–Hoeffding and Makarov bounds. Section 5 reports the robustness analysis, and Section 6 concludes.

2 The Identification of Treatment-Effect Heterogeneity

The three configurations that motivate this paper — marginals that leave heterogeneity undetermined, marginals that force it, and marginals that identify a common effect — are not independent examples. They are three readings of a single identification problem. This section states that problem as a proposition and derives the three cases as its corollaries, so that the reader meets one general principle rather than a list of illustrations.

2.1 The copula gap: what randomization does and does not identify

Fix a binary treatment $D \in \{0,1\}$ and potential outcomes $(Y(0), Y(1))$ with marginal CDFs $F_d(y) = P(Y(d) \leq y)$, $d \in \{0,1\}$, and joint CDF $H(y_1, y_0) = P(Y(1) \leq y_1, Y(0) \leq y_0)$. By Sklar's theorem (Sklar, 1959) there is a copula C with $H(y_1, y_0) = C(F_1(y_1), F_0(y_0))$: the copula carries the entire dependence structure between the two potential outcomes and is unique on $\text{Ran } F_1 \times \text{Ran } F_0$. Because the outcomes here are discrete, C is not unique off that grid, so we take the joint H as the primitive and write $\mathcal{H}(F_0, F_1)$ for the *Fréchet class* of all bivariate CDFs with marginals F_0, F_1 . For a functional $\theta(\cdot)$ the *identified set* is $\Theta_I(\theta) = \{\theta(H) : H \in \mathcal{H}(F_0, F_1)\}$.

Proposition 1 (Marginal identification and the copula gap). *Under random assignment (Assumption 1), F_0 and F_1 are point-identified, whereas the joint H is not: the distribution of the observed data (D, Y) depends on the law of $(Y(0), Y(1))$ only through $(F_0, F_1, P(D=1))$. Consequently the copula C is left unrestricted by the data — its identified set is the entire copula class compatible with (F_0, F_1) — and any parameter $\theta(H)$ that is not a functional of the marginals alone is at most partially identified, with $\Theta_I(\theta) = \{\theta(H) : H \in \mathcal{H}(F_0, F_1)\}$. The average treatment effect is the exception: $\tau = \int y dF_1(y) - \int y dF_0(y)$ is a marginal functional and is point-identified.*

Proof. Under Assumption 1, $Y \mid D=1 \sim F_1$ and $Y \mid D=0 \sim F_0$, so the law of the observables is the mixture $P(D=1) dF_1 + P(D=0) dF_0$: each unit reveals exactly one coordinate of $(Y(0), Y(1))$, and this law is invariant across all $H \in \mathcal{H}(F_0, F_1)$. Hence any two joints $H \neq H'$ in the Fréchet class are observationally equivalent, which gives point identification of F_0, F_1 and non-identification of H . For τ , linearity of expectation yields $\tau = E[Y(1)] - E[Y(0)] = \int y dF_1 - \int y dF_0$, a function of the marginals only. $\square$

Proposition 1 is the observation of Heckman, Smith and Clements (1997) that an experiment identifies the two marginals of the potential outcomes but not the distribution of their difference. Every object that describes heterogeneity is a functional of the unidentified H — equivalently, of the copula C :

- the distribution of individual effects $F_\Delta(\delta) = P(Y(1) - Y(0) \leq \delta) = \iint \mathbf{1}\{y_1 - y_0 \leq \delta\} dH(y_1, y_0)$;
- its dispersion $\text{Var}(\Delta) = \text{Var}(Y(1)) + \text{Var}(Y(0)) - 2 \text{Cov}(Y(1), Y(0))$, unidentified through the covariance $\text{Cov}(Y(1), Y(0)) = \iint [H - F_1 F_0] dy_1 dy_0$ (Hoeffding, 1940);
- the fraction made worse off $P(\Delta < 0)$, a functional of F_Δ ;
- rank preservation itself, which is the hypothesis $C = M$ that the copula equals the comonotone (upper Fréchet) bound.

Assuming a common effect, or rank preservation, is therefore not an innocuous simplification but the selection of one copula out of an unidentified class. This paper declines that selection and reports Θ_I instead.

2.2 The identified set and its three regimes

The endpoints of these identified sets are delivered by the sharp bounds of Section 4. The Fréchet–Hoeffding bounds (Proposition 2) confine H between the countermonotone lower bound $W = \max\{F_1 + F_0 - 1, 0\}$ and the comonotone upper bound $M = \min\{F_1, F_0\}$; the Makarov bounds (Proposition 3; Makarov 1981, Williamson and Downs 1990, Fan and Park 2010) sharpen this to pointwise bounds on F_Δ ; and the covariance is extremized at these same couplings — maximal at M , minimal at W — because the integrand $y_1 y_0$ is supermodular (Cambanis, Simons and Stout, 1976; Hoeffding, 1940). Each such $\theta(H) = \iint g dH$ is *linear* in H , and $\mathcal{H}(F_0, F_1)$ is convex and weakly compact, so every identified set $\Theta_I(\theta)$ is a compact interval whose endpoints are attained — a set, not a point.

Two facts about this interval organize the three cases.

Corollary 1 (Homogeneity is a location shift). *$\text{Var}(\Delta) = 0 \in \Theta_I(\text{Var } \Delta)$ if and only if $F_1(y) = F_0(y - \tau)$ for all y . Equivalently, a common additive effect $\Delta \equiv \delta$ (a.s.) is compatible with the marginals if and only if (F_0, F_1) lies in the location-shift family, in which case necessarily $\delta = \tau$.*

Proof. $\text{Var}(\Delta) = 0$ under some $H \in \mathcal{H}(F_0, F_1)$ iff Δ is a.s. constant under that H ; the constant must equal $E[\Delta] = \tau$; and $\Delta \equiv \tau$ a.s. iff $Y(1) = Y(0) + \tau$ a.s., i.e. $F_1(y) = P(Y(0) \leq y - \tau) = F_0(y - \tau)$ for all y . The comonotone coupling M realizes this degenerate Δ when the translate condition holds, so 0 is then the left endpoint of $\Theta_I(\text{Var } \Delta)$. $\square$

Lemma 1 (Degeneracy collapses the Fréchet class). *If F_0 is degenerate, $F_0 = \mathbf{1}\{\cdot \geq c_0\}$, then $\mathcal{H}(F_0, F_1)$ is the singleton $H(y_1, y_0) = F_1(y_1) \mathbf{1}\{y_0 \geq c_0\} = F_1(y_1) F_0(y_0)$, and the Fréchet–Hoeffding bounds coincide, $W \equiv M \equiv H$. Every functional of H is then point-identified; in particular $\Delta = Y(1) - c_0$ has identified law $F_\Delta(\delta) = F_1(\delta + c_0)$. If both marginals are degenerate then $\Delta = c_1 - c_0$ almost surely and $\text{Var}(\Delta) = 0$ is identified. The statement holds symmetrically when F_1 is degenerate.*

Proof. With $Y(0) = c_0$ a.s., $H(y_1, y_0) = F_1(y_1)$ for $y_0 \geq c_0$ and $H = 0$ for $y_0 < c_0$; since $F_0(y_0) = \mathbf{1}\{y_0 \geq c_0\} \in \{0, 1\}$, this equals $F_1(y_1) F_0(y_0)$, the independence coupling. Substituting $F_0 \in \{0, 1\}$ into $M = \min\{F_1, F_0\}$ and $W = \max\{F_1 + F_0 - 1, 0\}$ gives $M = W = F_1 F_0$. A degenerate random variable is independent of every random variable, so the copula plays no role and $\mathcal{H}(F_0, F_1)$ contains this single element; the change of variables $\Delta = Y(1) - c_0$, and $\Delta = c_1 - c_0$ when $F_1 = \mathbf{1}\{\cdot \geq c_1\}$, give the remaining claims. $\square$

Corollary 1 and Lemma 1 partition the marginals into three regimes: equal and non-degenerate (0 interior to a wide interval, Case 1), not a location shift (0 excluded, Case 2), and degenerate (interval a singleton, Case 3). The binary specialization used below relies on one sharp interval. Writing the joint of the binary pair $(Y(1), Y(0))$ as the 2×2 table with cells $a = P(1, 1)$, $\pi_+ =$

$P(Y(1)=1, Y(0)=0)$, $\pi_- = P(Y(1)=0, Y(0)=1)$, $b = P(0,0)$ and margins $p_d = P(Y(d)=1)$, the constraints $a + \pi_+ = p_1$, $a + \pi_- = p_0$, and $a, b, \pi_{\pm} \geq 0$ leave a single free parameter, giving the sharp identified set

$$\pi_- \in [\max\{0, p_0 - p_1\}, \min\{p_0, 1 - p_1\}], \quad \pi_+ - \pi_- = p_1 - p_0 = \tau, \quad (1)$$

every value of which is attained by a valid table; this is the Boole–Fréchet bound formalized in Corollary 2, with π_- the fraction *made worse off*.

2.3 Case 1: identical non-degenerate marginals — a wide identified set

Let $F_0 = F_1$ be non-degenerate; the binary instance is $F_0 = F_1 = \text{Bernoulli}(\frac{1}{2})$, so $p_0 = p_1 = \frac{1}{2}$ and $\tau = 0$. Substituting into (1), the harmed fraction is set-identified as $\pi_- \in [0, \frac{1}{2}]$, and the two endpoints are attained at the extremal copulas: the comonotone M gives $\pi_- = 0$ with $\Delta \equiv 0$, and the countermonotone W gives $\pi_- = \frac{1}{2}$ with $\Delta = \pm 1$ each with probability $\frac{1}{2}$; correspondingly $\text{Var}(\Delta) \in [0, 1]$. The table below displays a ten-unit realization of each.

Units 1-7	Comonotone M : common effect			Countermonotone W : maximal heterogeneity		
	$Y_i(0)$	$Y_i(1)$	Δ_i	$Y_i(0)$	$Y_i(1)$	Δ_i
1-5	0	0	0	0	1	+1
6-10	1	1	0	1	0	-1
3-10	Rank Invariance			can I rule out heterogeneous treatment effect		
Each arm	5 employed, 5 not			5 employed, 5 not		
	how should I rule out the common effect					

Table 1: Two extremal couplings of the same marginals $F_0 = F_1 = \text{Bernoulli}(\frac{1}{2})$ ($Y = 1$ for employed). Both reproduce exactly five employed and five not in *each* arm — all the experiment observes — so $p_0 = p_1 = \frac{1}{2}$ and $\tau = 0$ in both. Under the comonotone coupling M the effect is common and zero; under the countermonotone coupling W every unit is moved, half helped and half harmed. The independence coupling Π produces the same margins again.

By Proposition 1 the couplings M , W , and independence Π are observationally equivalent: all lie in $\mathcal{H}(F_0, F_1)$, so the treated and control samples are drawn from F_1 and F_0 whichever holds, and the likelihood of the observed data is identical across them. Homogeneity ($\text{Var } \Delta = 0$) belongs to the identified set — by Corollary 1, since $F_1 = F_0 = F_0(\cdot - 0)$ with $\tau = 0$ — but so does maximal heterogeneity ($\text{Var } \Delta = 1$). The experiment cannot tell them apart; heterogeneity is *possible* but *not forced*.

2.4 Case 2: marginals that are not a translate — homogeneity excluded

Now suppose a job-training program raises *average* earnings by \$1,000, while the share earning *zero* rises from 20% among controls to 30% among the treated; earnings are non-negative. Two logically independent arguments each place the entire identified set in the heterogeneous region.

Part A (marginal-only; via Corollary 1). The mean rose, so $\tau = E[Y(1)] - E[Y(0)] = 1000 > 0$. Under a common effect, Corollary 1 forces $\Delta \equiv \tau$ a.s., i.e. $Y(1) = Y(0) + 1000$; then $\{Y(1) =$

$0\} = \{Y(0) = -1000\}$, which has probability zero because earnings are non-negative, contradicting $P(Y(1) = 0) = 0.30$. Hence (F_0, F_1) is not a location shift, and by Corollary 1, $0 \notin \Theta_I(\text{Var } \Delta)$; $\text{Var}(\Delta) > 0$ under *every* coupling. This uses only the marginals and the non-negativity of the support.

Part B (joint; via the Fréchet floor). Coarsen the outcome to the employment indicator $E = \mathbf{1}\{Y > 0\}$, so $p_0 = P(Y(0) > 0) = 0.80$ and $p_1 = 0.70$. Then (1) gives $\pi_- \in [\max\{0, 0.10\}, \min\{0.80, 0.30\}] = [0.10, 0.30]$: under *every* coupling at least 10% of units lose employment, even though average earnings rose. This uses only the two employment rates and the Fréchet lower bound, not the argument of Part A.

The two arguments are distinct and non-redundant. Part A establishes positive dispersion, $\text{Var}(\Delta) > 0$, from the shape of the earnings marginals; Part B establishes a strictly positive mass of *losers*, $\pi_- \geq 0.10$, from the employment rates. Neither implies the other — effects can be dispersed yet all non-negative (no losers), and a positive harmed fraction is a statement about the sign of individual effects rather than their spread — and Case 2 exhibits both. Heterogeneity is not assumed but *forced*, and, on the employment margin, so is harm.

2.5 Case 3: degenerate marginals — the identified set collapses

The exception is degeneracy. By Lemma 1, if every control unit shares the outcome $Y(0) \equiv c_0$ then $\mathcal{H}(F_0, F_1)$ is a singleton, the copula is irrelevant, and F_Δ is point-identified from the treated marginal alone; the collapse is exact, $W \equiv M \equiv H$. A common effect is then *identified*, not merely assumed, precisely in the doubly degenerate case $Y(0) \equiv c_0$, $Y(1) \equiv c_1$, where $\Delta = c_1 - c_0$ for every unit and $\text{Var}(\Delta) = 0$. Near-degeneracy is the empirical version: if 98% graduate whether treated or not, $p_0 = p_1 = 0.98$ and (1) traps the harmed fraction in $[0, 0.02]$; as $p_0, p_1 \rightarrow 1$ the interval shrinks to $\{0\}$. Case 3 is therefore the degenerate limit of Case 1 — the same identical-marginals configuration driven to the boundary at which the Fréchet class contracts to a point. Identification of a common effect comes from the *absence of dispersion* in the potential outcomes, i.e. from ceilings and floors, not from any assumption on dependence.

2.6 Synthesis: three locations of one identified set

The three cases are three positions of the same object, the identified set $\Theta_I(\text{Var } \Delta)$ implied by the marginals.

Case	Marginal configuration	$\Theta_I(\text{Var } \Delta)$	Common-effects hypothesis
1	$F_0 = F_1$, non-degenerate	$[0, \bar{V}]$, $\bar{V} > 0$	neither refuted nor confirmed
2	not a location shift	$[\underline{V}, \bar{V}]$, $\underline{V} > 0$	refuted
3	F_0 or F_1 (near-)degenerate	$\{0\}$ (in the limit)	confirmed

Table 2: The three cases as three locations of one identified set. Reading down the third column, $\Theta_I(\text{Var } \Delta)$ is wide and anchored at zero (Case 1), lifts off zero (Case 2), and collapses onto zero as a singleton (Case 3). The status of the common-effects hypothesis in the fourth column is read directly from where zero sits relative to the identified set.

The experiment never reveals *where* in the interval the truth lies — that is the content of Proposition 1 — but it does reveal the interval, and the interval alone decides whether a common-effects report is undetermined (Case 1), false (Case 2), or (near-)true (Case 3). Section 4 develops the Fréchet–Hoeffding and Makarov bounds that produce these intervals, and Section 5 turns their width into the paper’s robustness metric.

3 Minjae’s Framework

3.1 Setup and notation

Consider a binary treatment $D \in \{0, 1\}$ and a discrete outcome Y . Following the potential-outcomes notation, let $Y_i(0)$ and $Y_i(1)$ denote the potential outcomes of unit i under control and treatment. Only one of the two is ever realized: the observed outcome is

$$Y_i = Y_i(0) + (Y_i(1) - Y_i(0)) D_i = D_i Y_i(1) + (1 - D_i) Y_i(0).$$

The *individual treatment effect* is $\Delta_i \equiv Y_i(1) - Y_i(0)$, and the average treatment effect is $\tau \equiv E[\Delta_i] = E[Y_i(1) - Y_i(0)]$. We maintain two assumptions throughout.

Assumption 1 (Random assignment). $D_i \perp (Y_i(0), Y_i(1))$.

Assumption 2 (SUTVA). There is a single, well-defined version of each potential outcome, and no interference across units: unit i ’s outcome depends only on its own treatment D_i .

Assumption 1 holds by design in a completely randomized experiment and lets us proceed without modeling selection. Under Assumptions 1–2 the two *marginal* distributions

$$F_1(y) \equiv P(Y_i(1) \leq y), \quad F_0(y) \equiv P(Y_i(0) \leq y),$$

are point-identified from the treated and control subsamples. The *joint* distribution $H(y_1, y_0) \equiv P(Y_i(1) \leq y_1, Y_i(0) \leq y_0)$ is not: we never observe $Y_i(1)$ and $Y_i(0)$ for the same unit. This is the fundamental problem of causal inference, and it is the source of every identification difficulty in what follows. Because Y is discrete, F_1 and F_0 have jumps; throughout, $F_d^{-1}(\nu) = \inf\{y : F_d(y) \geq \nu\}$

denotes the generalized inverse (quantile function), the object that the quantile treatment effect and the comonotone coupling below rely on.

3.2 Two decompositions of the treatment effect

It is useful to separate two distinct ways of decomposing τ , because they answer different questions and only one of them speaks to heterogeneity.

(A) Decomposition by treatment status.

By the law of total expectation,

$$\tau = E[\Delta_i] = \underbrace{E[\Delta_i \mid D_i = 1]}_{\text{ATET}} P(D_i = 1) + \underbrace{E[\Delta_i \mid D_i = 0]}_{\text{ATEU}} P(D_i = 0). \quad (2)$$

This is an aggregation identity over treatment *status*: it splits the ATE into the average effect on the treated (ATET) and on the untreated (ATEU). It is *not* a separation into a homogeneous and a heterogeneous part. In fact, under Assumption 1 we have $\text{ATET} = \text{ATEU} = \tau$, so (2) conveys nothing about whether effects vary across units. We record it only to distinguish it from the decomposition we actually use.

(B) Common/differential decomposition.

Write the individual effect as

$$\Delta_i = \underbrace{\tau}_{\text{common}} + \underbrace{u_i}_{\text{differential}}, \quad u_i \equiv \Delta_i - \tau, \quad E[u_i] = 0. \quad (3)$$

Here τ is the part of the effect shared by every unit and u_i is the unit-specific (idiosyncratic) departure from it. In regression form this is the representation of Djebbari and Smith (2008):

$$Y_i = \beta_0 + \tau D_i + [u_i D_i + \varepsilon_i], \quad \beta_0 = E[Y_i(0)], \quad \varepsilon_i = Y_i(0) - E[Y_i(0)]. \quad (4)$$

Conditioning on covariates X_i refines (3) into a *systematic* component $\tau(X_i) = \tau_0 + X_i' \tau_X$, which varies across observable subgroups and is point-identified from treatment-covariate interactions, and a residual *idiosyncratic* component $u_i - (\tau(X_i) - \tau)$, which varies within subgroups and is not. The common-effects benchmark is the special case $u_i \equiv 0$ ($\Delta_i = \tau$ for all i); the heterogeneous-effects model relaxes it. The object this paper targets is therefore the *distribution of the differential component* u_i — in particular its dispersion and its left tail.

Why the differential component is not point-identified.

The decomposition (3) is just an accounting identity; the difficulty is that the law of u_i depends on the unidentified joint H . Concretely,

$$\text{Var}(\Delta_i) = \text{Var}(Y_i(1)) + \text{Var}(Y_i(0)) - 2 \text{Cov}(Y_i(1), Y_i(0)). \quad (5)$$

A randomized experiment identifies the two marginal variances but says nothing about $\text{Cov}(Y_i(1), Y_i(0))$, which ranges over an interval as the unknown dependence between the potential outcomes varies. The same logic applies to the tail probability $P(\Delta_i < 0)$ — the fraction of units made *worse off* by treatment — and to the quantiles of Δ_i . These policy-relevant features of heterogeneity are point-identified only if one is willing to fix the dependence structure. This paper declines to do so, and instead characterizes the identified set.

3.3 Four parameters for impact heterogeneity

The literature offers four distinct objects for going beyond the mean. They differ in what they identify, in the primitive and the assumption each requires, and in the policy question each answers. The table below contrasts them; the text then states which this paper adopts and why.

Parameter	Object	Identifying requirement	Policy question
Fréchet–Hoeffding bounds	$\max\{0, F_1 + F_0 - 1\} \leq H \leq \min\{F_1, F_0\}$	The two marginals F_1, F_0 ; <i>no</i> dependence assumption	Worst-case dispersion and fraction made worse off
Partial identification	Identified set Θ_I of values consistent with data and maintained assumptions	Credible restrictions only; <i>no</i> point-identifying assumption	A robust interval for any functional of the joint law
Quantile treatment effect	$\text{QTE}(\nu) = F_1^{-1}(\nu) - F_0^{-1}(\nu)$	Marginals only; <i>rank preservation</i> to read as individual effects	How treatment reshapes the outcome distribution (inequality)
Marginal treatment effect	$\text{MTE}(x, u) = E[\Delta_i \mid X=x, U_D=u]$	A continuous instrument with support, plus monotonicity	Effect of <i>expanding</i> the program at the margin (PRTE)

Table 3: Four parameters for impact heterogeneity. H is the joint distribution of $(Y_i(1), Y_i(0))$, $\Delta_i = Y_i(1) - Y_i(0)$, and U_D is unobserved resistance to treatment. The four sit on a ladder of assumptions: the QTE needs only the marginals, the Fréchet–Hoeffding bounds add no dependence assumption, and the MTE requires an instrument and a selection model.

Fréchet–Hoeffding bounds.

Given the two marginals, the joint H is confined between the comonotone coupling — perfect positive dependence, i.e. rank preservation, the upper bound — and the countermonotone coupling, the lower bound. Propagating this constraint to the distribution of Δ_i yields sharp bounds on $\text{Var}(\Delta_i)$ and on the fraction harmed $P(\Delta_i < 0)$; for binary outcomes these reduce to the “mover” probabilities reported by Djebbari and Smith (2008). The defining virtue is that the bound requires *only the two marginals* and no assumption on dependence. Section 4 develops it formally.

Partial identification.

The bounds above are one instance of the broader partial-identification program (Manski): when the data plus credible assumptions do not pin down a single value, report the entire identified set Θ_I rather than impose an incredible assumption to force a point. Here the marginals are point-identified but every functional of the joint — $\text{Var}(\Delta_i)$, quantiles of Δ_i , the fraction harmed — is only set-identified. The stance is deliberate: we prefer a credible interval to an incredible number.

Quantile treatment effect.

The QTE compares the two marginals rank by rank,

$$\text{QTE}(\nu) = F_1^{-1}(\nu) - F_0^{-1}(\nu), \quad \nu \in (0, 1), \quad (6)$$

and is point-identified from the marginals alone — no joint required. Its interpretation, however, must be stated carefully. The QTE is the effect *at* rank ν of the distribution, a statement about how treatment reshapes the outcome distribution (and thus about inequality). It equals the effect *on* the particular unit sitting at rank ν only under rank preservation (the comonotone case above); absent that assumption the unit at rank ν in the treated world need not be the same unit as at rank ν in the control world. This object is also the bridge to the testing literature: the constant-effect null of Chung and Olivares (2021), $H_0 : F_1(y + \delta) = F_0(y) \ \forall y$, is equivalent via the Doksum–Lehmann model (their Remark~2) to $\text{QTE}(\nu) = \delta$ for all ν — a *flat* QTE function. Their contribution is an asymptotically valid permutation test of this null that remains pivotal despite the estimation of the nuisance shift $\hat{\delta} = \bar{Y}_1 - \bar{Y}_0$, achieved through a Khmaladze martingale transformation of the empirical process. In our framework the QTE is reported as a distributional summary; we do *not* read it as a vector of individual effects, precisely because we decline rank preservation.

Marginal treatment effect.

The MTE (Heckman–Vytlacil) is a selection-model object. It posits a choice equation $D_i = \mathbf{1}\{p(Z_i) \geq U_{D,i}\}$, where $U_{D,i} \sim \text{Unif}[0, 1]$ is the unit’s unobserved resistance to treatment and $p(Z_i)$ is the propensity score, and defines

$$\text{MTE}(x, u) = E[\Delta_i \mid X_i = x, U_{D,i} = u]. \quad (7)$$

It is a “mother parameter” in the sense that every conventional estimand is a weighted integral of it — $\tau = \int_0^1 \text{MTE}(x, u) du$ for the ATE, and analogous weightings for ATT, LATE, and the policy-relevant treatment effect (PRTE) of a marginal program expansion. Identifying (7) requires a *continuous instrument* with sufficient support to trace $p(Z)$ across $[0, 1]$, together with monotonicity. This is the crucial point of contrast with the present paper: a completely randomized design with full compliance sets $Z_i = D_i$ and $p(Z_i) \in \{0, 1\}$, so there is no interior margin of indifference and no selection on gains to recover. The MTE curve is not traced out; it collapses to the single point τ . The MTE answers a different, choice-theoretic question — who selects into treatment on the basis of their own gains, and what happens when the program is expanded at the margin — that lies outside the assumptions maintained here. We therefore treat it as a *contrastive* framework rather

than a tool of this paper.

3.4 The framework adopted in this paper

The four objects are not interchangeable; they sit on a ladder of assumptions. The QTE is the cheapest — point-identified from the marginals — but is silent about the joint distribution and, without rank preservation, cannot be read as individual heterogeneity. The MTE is the richest, but it is built for a selection problem this design eliminates and collapses to the ATE under full randomization. Between them, the Fréchet–Hoeffding/Makarov bounds occupy the position this paper adopts: they take the two randomization-identified marginals as primitives, impose no dependence structure — in particular *no* rank preservation — and return the identified set for the distribution of the differential component u_i , including $\text{Var}(\Delta_i)$ and the fraction harmed $P(\Delta_i < 0)$. The QTE is reported alongside as a distributional summary, and the MTE is invoked only to mark the boundary of the design. Section 4 develops these bounds; Section 5 uses them to assess how far estimated effects can move once the common-effects assumption is relaxed.

4 Partial Identification and Fréchet–Hoeffding Bounds

A randomized experiment delivers the two marginals F_1, F_0 but never the joint law H of $(Y_i(1), Y_i(0))$. We therefore proceed in three steps: bound the joint given the marginals (Fréchet–Hoeffding), translate that into sharp bounds on the distribution of the effect Δ_i (Makarov), and read off the identified set for the policy-relevant features of heterogeneity — the fraction made worse off and the dispersion of effects.

4.1 Bounding the joint distribution

Proposition 2 (Fréchet–Hoeffding bounds). *Let $(Y_i(1), Y_i(0))$ have marginal CDFs F_1, F_0 and joint CDF H . For every $(y_1, y_0) \in \mathbb{R}^2$,*

$$\underbrace{\max\{0, F_1(y_1) + F_0(y_0) - 1\}}_{W(y_1, y_0)} \leq H(y_1, y_0) \leq \underbrace{\min\{F_1(y_1), F_0(y_0)\}}_{M(y_1, y_0)}. \quad (8)$$

Both W and M are themselves bivariate CDFs with marginals F_1, F_0 , so neither bound can be tightened without further assumptions. The upper bound M is attained under comonotonicity (perfect positive dependence, i.e. rank preservation) and the lower bound W under countermonotonicity.

Proof. Let $A = \{Y_i(1) \leq y_1\}$ and $B = \{Y_i(0) \leq y_0\}$, so $H(y_1, y_0) = P(A \cap B)$. Since $A \cap B \subseteq A$ and $A \cap B \subseteq B$, $P(A \cap B) \leq \min\{P(A), P(B)\} = M$. Since $P(A \cap B) = P(A) + P(B) - P(A \cup B) \geq P(A) + P(B) - 1$ and $P(A \cap B) \geq 0$, we obtain $H \geq W$. Sharpness follows from Sklar’s theorem: the Fréchet–Hoeffding copulas $M(u, v) = \min\{u, v\}$ and $W(u, v) = \max\{0, u + v - 1\}$ are themselves copulas, realized by the comonotone pair $(F_1^{-1}(U), F_0^{-1}(U))$ and the countermonotone pair $(F_1^{-1}(U), F_0^{-1}(1 - U))$ with $U \sim \text{Unif}[0, 1]$. $\square$

4.2 Sharp bounds on the distribution of effects

The marginals do not pin down the law of $\Delta_i = Y_i(1) - Y_i(0)$, but they bound it sharply. The bounds below are the distributional analogue of (8).

Proposition 3 (Makarov bounds). *The CDF $F_\Delta(\delta) = P(\Delta_i \leq \delta)$ satisfies, for every δ , $F_\Delta^L(\delta) \leq F_\Delta(\delta) \leq F_\Delta^U(\delta)$, where*

$$F_\Delta^L(\delta) = \sup_y \max\{F_1(y) - F_0(y - \delta), 0\}, \quad F_\Delta^U(\delta) = 1 + \inf_y \min\{F_1(y) - F_0(y - \delta), 0\}. \quad (9)$$

These bounds are pointwise sharp and are attained at dependence structures consistent with F_1, F_0 .

The result is due to Makarov (1981) and Williamson and Downs (1990); see Fan and Park (2010) for the treatment-effect formulation in (9). It is the engine of the paper: every quantity we report below is a functional of F_Δ , so each inherits an identified set from (9). Unlike the quantile treatment effect of (6), which describes the *marginals*, F_Δ describes the *distribution of individual effects* and is genuinely set-identified.

Two points sharpen the reading of (9). First, F_Δ^L and F_Δ^U are each valid CDFs and each is sharp *pointwise*, but they are not jointly attained by a single coupling; the sharp identified set for a functional of F_Δ (a quantile, the variance) is obtained by optimizing that functional over all CDFs F with $F_\Delta^L \leq F \leq F_\Delta^U$, which is why we treat the variance separately in (11) rather than reading it off the envelope. Second, the bounds carry no information about the *mean*: integrating gives $E[\Delta_i] = E[Y_i(1)] - E[Y_i(0)] = \tau$ regardless of the coupling, so it is precisely the spread and the tails of F_Δ — not its center — that the joint leaves undetermined.

4.3 The fraction made worse off

For the discrete outcomes that motivate this paper, the bounds take a transparent form.

Corollary 2 (Binary outcomes). *Let $Y \in \{0, 1\}$ and $p_d = P(Y_i(d) = 1)$, $d \in \{0, 1\}$. Then $\Delta_i \in \{-1, 0, 1\}$, the fraction made worse off $\pi_- = P(\Delta_i = -1) = P(Y_i(1) = 0, Y_i(0) = 1)$ satisfies*

$$\max\{0, p_0 - p_1\} \leq \pi_- \leq \min\{p_0, 1 - p_1\}, \quad (10)$$

and the fraction helped $\pi_+ = P(\Delta_i = 1)$ satisfies $\max\{0, p_1 - p_0\} \leq \pi_+ \leq \min\{p_1, 1 - p_0\}$.

Proof. Apply the Boole–Fréchet inequalities to the event $\{Y_i(1) = 0\} \cap \{Y_i(0) = 1\}$: $P(\cdot) \geq P(Y_i(0) = 1) - P(Y_i(1) = 1) = p_0 - p_1$ and $P(\cdot) \geq 0$ give the lower bound, while $P(\cdot) \leq \min\{P(Y_i(0) = 1), P(Y_i(1) = 0)\} = \min\{p_0, 1 - p_1\}$ gives the upper bound. The argument for π_+ is symmetric. $\square$

The average effect $\tau = p_1 - p_0 = \pi_+ - \pi_-$ is point-identified, but its split into *winners* and *losers* is not: (10) is exactly the bound on the **mover** probabilities reported by Djebbari and Smith (2008). The policy content is direct — a program with $\tau > 0$ on average is consistent with anywhere from no one to $\min\{p_0, 1 - p_1\}$ of the population being harmed, and only the experiment’s marginals, not an unjustified dependence assumption, should determine which.

4.4 The identified set for the dispersion of effects

Corollary 3 (Dispersion and forced heterogeneity). *By Hoeffding’s identity $\text{Cov}(Y_i(1), Y_i(0)) = \iint [H(y_1, y_0) - F_1(y_1)F_0(y_0)] dy_1 dy_0$ and (8), the covariance is interval-identified, with its maximum at the comonotone coupling and its minimum at the countermonotone coupling. Hence*

$$\text{Var}(\Delta_i) = \text{Var}(Y_i(1)) + \text{Var}(Y_i(0)) - 2 \text{Cov}(Y_i(1), Y_i(0)) \quad (11)$$

is interval-identified, its lower endpoint corresponding to the comonotone (rank-preserving) coupling and its upper endpoint to the countermonotone coupling. The common-effects value $\text{Var}(\Delta_i) = 0$ belongs to this set only if the marginals are pure translates, $F_1(y) = F_0(y - \tau)$ for all y .

The last clause is the paper’s identifying message. If the treated and control marginals are *not* a location shift of one another — as is typically the case once the outcome is discrete, since a shift would have to map the support of F_0 onto that of F_1 — then Δ_i cannot be almost surely constant. Heterogeneity is then not a modeling option but a logical consequence of the marginals the experiment identifies, and the only open question is its magnitude, which (9) and (11) bound.

This magnitude is explicit. The comonotone coupling gives the maximal covariance $\text{Cov}_{\max} = \iint [\min\{F_1(y_1), F_0(y_0)\} - F_1(y_1)F_0(y_0)] dy_1 dy_0$, so the dispersion of effects is bounded below by

$$\underline{V} \equiv \text{Var}(Y_i(1)) + \text{Var}(Y_i(0)) - 2 \text{Cov}_{\max} \geq 0, \quad (12)$$

with equality if and only if $F_1(y) = F_0(y - \tau)$. For the binary case of Corollary 2 this lower bound has a clean closed form: the comonotone coupling sends the harmed fraction to zero ($\pi_- = 0$) and leaves $\Delta_i \in \{0, 1\}$ with $P(\Delta_i = 1) = |\tau|$, so

$$\underline{V} = |\tau| (1 - |\tau|). \quad (13)$$

Unless the average effect is exactly zero, then, the variance of individual effects is bounded away from zero even under the most favorable (rank-preserving) dependence: a nonzero mean impact on a binary outcome *mechanically* forces heterogeneous individual effects.

Remark 1 (No rank preservation is imposed). The comonotone (upper) bound in (8) is perfect positive dependence: it would point-identify the effect as $\Delta_i = \text{QTE}(F_0(Y_i(0)))$ and drive the fraction harmed to its lower bound. We do not impose it. Working with the full Fréchet–Hoeffding and Makarov bounds reports the identified set across *all* dependence structures, so the conclusions are robust to arbitrary reordering of units between the treated and control distributions.

A caveat on the maintained assumptions.

The bounds rest on Assumptions 1–2. SUTVA — the *Stable Unit Treatment Value Assumption* — rules out interference and multiple versions of treatment; under spillovers (for instance, the general-equilibrium effects of a large transfer program) it can fail, and F_1, F_0 would lose their clean potential-outcome interpretation. We maintain it throughout. Because assignment is randomized, no selection model is needed: the marginals are identified directly from the treated and control subsamples, which is exactly what (8)–(9) take as primitives.

5 Robustness

We now ask the question that motivates the paper: how much can a treatment effect that is reported under the common-effects assumption be trusted once that assumption is relaxed? The bounds of Section 4 turn this into a measurable exercise rather than a rhetorical one. Fix a policy-relevant target θ — the fraction made worse off π_- , the dispersion $\text{Var}(\Delta_i)$, or a quantile of F_Δ . Each is a functional of the joint law H , and by Propositions 2–3 the data identify only its *robustness region*

$$\Theta_I(\theta) = [\theta^L, \theta^U], \quad (14)$$

the set of values consistent with the experiment’s marginals across all dependence structures. The common-effects benchmark reports a single point θ^{CE} inside (or, revealingly, outside) this region.

A departure metric.

We summarize robustness by how far the common-effects report can be from the truth in the worst case,

$$R(\theta) = \frac{\theta^U - \theta^L}{|\theta^{\text{CE}}| \vee 1}, \quad (15)$$

the width of the identified set relative to the reported magnitude. A small $R(\theta)$ means the common-effects number is robust: even under the most adverse reordering of units, the policy conclusion (say, “fewer than five percent are harmed”) survives. A large $R(\theta)$ means the reported point is an artifact of the homogeneity assumption and the honest object is the interval (14). The two endpoints have clean interpretations from Section 4: θ^L is the comonotone, rank-preserving case that minimizes apparent heterogeneity, and θ^U is the countermonotone case that maximizes it.

What the exercise shows.

Two features of the bounds discipline the reading. First, by Corollary 3, whenever the treated and control marginals are not a pure translate the benchmark $\text{Var}(\Delta_i) = 0$ falls *outside* Θ_I , so R is undefined in the favorable direction — homogeneity is rejected by the marginals alone, before any dependence assumption. Second, by Corollary 2, the fraction harmed is bounded below by $\max\{0, p_0 - p_1\}$, so a program that raises the outcome on average ($p_1 > p_0$) is consistent with *zero* losers but does not require it; the upper bound $\min\{p_0, 1 - p_1\}$ measures how much worse the distributional story could be than the mean suggests. Consistent with this logic, Djebbari and Smith (2008) find for PROGRESA that only a small fraction of households are made worse off — an empirical instance in which the robustness region is narrow enough to support the policy conclusion.

6 Conclusion

This paper separated a treatment effect into a common component, shared by every unit, and a differential component that varies across them, and asked what a randomized experiment can

say about the second once the convenient assumption of homogeneity is dropped. The answer is a partial-identification one. The experiment delivers the two marginal distributions of potential outcomes but not their joint law, so the distribution of individual effects — and with it the fraction made worse off and the dispersion of effects — is set-identified rather than point-identified. The Fréchet–Hoeffding and Makarov bounds characterize that set sharply, without any assumption on the dependence between potential outcomes and, in particular, without rank preservation.

Two messages follow. Methodologically, the four parameters surveyed in Section 3 are not substitutes: the quantile treatment effect describes the marginals and reads as individual heterogeneity only under rank preservation; the marginal treatment effect answers a selection question that a fully randomized design sets aside; and the bounds occupy the middle ground this design actually supports. Substantively, heterogeneity need not be assumed to be found — when the marginals are not a translate of one another it is forced by the data, and the only question the analyst can honestly answer is how large it is, which the bounds quantify. Reporting the robustness region (14) alongside the headline effect is, on this view, not a robustness afterthought but the policy-relevant object itself.

A Reproducing the Identified Sets

Every identified set reported in Section 2 is computed from the two marginals alone by the replication (base R, no external data). The script follows the identification logic of Section 2.1 directly: because a randomized experiment fixes only F_0 and F_1 , the whole exercise is to sweep the Fréchet class $\mathcal{H}(F_0, F_1)$ and record how far each target can travel. It writes the two tables and the figure reproduced below.

A.1 Construction

The script builds three objects, each a direct implementation of a result in Section 4.

1. **Extremal couplings.** On $[0, 1]$ each outcome y_j occupies a quantile slice of F_1 and of F_0 ; overlapping the slices with the two uniforms aligned yields the comonotone coupling M , and with them reflected ($U \leftrightarrow 1 - U$) the countermonotone coupling W (Proposition 2). This is the entire body of `coupling(): J[j,k] = overlap(A_lo[j], A_hi[j], B_lo[k], B_hi[k])`.
2. **The dispersion set.** From a coupling J the script forms $\text{Cov}(Y(1), Y(0))$ and hence $\text{Var}(\Delta)$; by Cambanis, Simons and Stout (1976) the covariance is maximal at M and minimal at W , so $[\text{Var}_M(\Delta), \text{Var}_W(\Delta)]$ is the sharp set of Corollary 3.
3. **The harmed-fraction set.** For binary margins the closed form (1) is applied directly and cross-checked against the Makarov bounds (Proposition 3) evaluated just below zero; the two agree to machine precision in every case.

Running the script confirms, through `stopifnot` assertions, that each computed set equals the value stated in Section 2.

A.2 The Case 2 marginals

The discrete earnings marginals behind Section 2.4 are given below (support in dollars). They raise mean earnings by exactly \$1,000 while lifting the mass at zero from 0.20 to 0.30; by Corollary 1 no location shift can do both, so $\text{Var}(\Delta) = 0$ is excluded before any dependence assumption is entertained.

Earnings (\$)	0	1,000	2,000	3,000	4,000	5,000	Mean
Control F_0	0.20	0.30	0.25	0.15	0.07	0.03	1,680
Treated F_1	0.30	0.04	0.06	0.10	0.28	0.22	2,680

Table 4: The discrete earnings marginals used in Case 2 (probability mass at each support point). Mean earnings rise by \$1,000 while the mass at zero rises from 0.20 to 0.30; no common additive effect can produce both, so heterogeneity is forced.

A.3 Computed identified sets

The table collects the sets the script returns; the figure plots the harmed-fraction sets on the common, unit-free $[0, 1]$ scale.

Case	Configuration	$\Theta_I(\text{Var } \Delta)$	$\Theta_I(\pi_-)$	Common effect
1	$F_0 = F_1$, Bernoulli($\frac{1}{2}$)	$[0, 1]$	$[0, 0.50]$	undetermined
2	earnings, not a translate	$[1.14, 10.4] \times 10^6$	$[0.10, 0.30]$	refuted
3	ceiling, $p = 0.98$	$[0, 0.04]$	$[0, 0.02]$	(near-)confirmed

Table 5: Identified sets computed by `scripts/fh_makarov_bounds.R`. The variance sets are in the outcome’s own units (Case 2 in dollars²; Cases 1 and 3 on $\{0, 1\}$), so only π_- , which is unit-free, is compared across rows in the figure. For Case 2, π_- is read on the employment margin $\mathbf{1}\{Y > 0\}$.

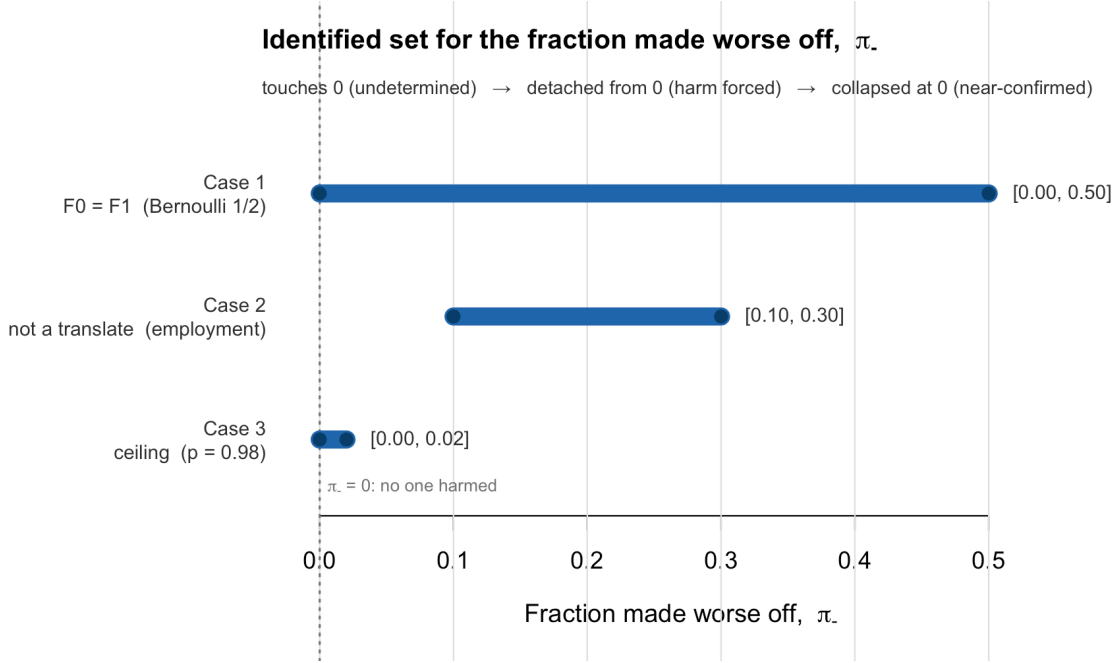


Figure 1: Identified sets for the fraction made worse off π_- across the three base cases, computed from the marginals alone. The dotted line $\pi_- = 0$ is the no-harm benchmark implied by a beneficial common effect. Reading top to bottom, the set touches the benchmark (Case 1, undetermined), detaches from it (Case 2, harm forced), and collapses onto it (Case 3, near-confirmed).

How to read the figure, and why it matters.

Each bar is the identified set $\Theta_I(\pi_-)$ for the fraction made worse off; the dotted line at $\pi_- = 0$ is the benchmark that a beneficial *common* effect would imply — no one harmed. The three bars make the regimes of Section 2.6 quantitative. In Case 1 the bar spans $[0, 0.50]$ and *touches* the benchmark: the data are equally consistent with no one harmed and with half the population harmed, so the common-effects reading is undetermined. In Case 2 the bar $[0.10, 0.30]$ is *detached* from the benchmark: every dependence structure leaves at least ten percent worse off, so the no-harm reading is refuted by the marginals themselves. In Case 3 the bar $[0, 0.02]$ has *collapsed* onto the benchmark: harm is at most two percent, and a near-common effect is confirmed by the ceiling rather than assumed.

Application.

The width of each bar is exactly the numerator $\theta^U - \theta^L$ of the robustness metric $R(\theta)$ in (15). Reporting it beside a headline effect converts the question “is the common-effects number safe?” into a measurement: a bar that is narrow and far from the benchmark (Case 3) certifies the headline; a bar that straddles it (Case 1) warns that the headline is an artifact of the homogeneity assumption; and a bar detached from it (Case 2) shows the headline is refuted outright. The three functions that produced this figure apply unchanged to estimated marginals $\hat{F}_1, \hat{F}_0$, which is how the empirical robustness table anticipated in Section 5 would be populated.